Perry–Beurling Spectral Sieve: Status Note on Diagnostic Lemmas and Grand Arithmetic Campaign

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Private research note — not a proof of RH

`luckyseoul/perry-beurling-spectral-sieve`

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Abstract

We summarize the reconstructed Perry–Beurling Spectral Sieve (PBSS): a projection diagnostic for density residuals on log-windows, with model theorems A_0/B_0 proved as lemmas M1–M4, and a grand numerical campaign of the arithmetic Chebyshev residual through $x_{\max} = 10^{10}$. The instrument behaves as designed on critical-line and defect controls. The arithmetic residual soft-plateaus at $R_d \approx 0.15$–0.19 and does *not* exhibit A_0-style collapse. Full Theorems A and B, and the Riemann Hypothesis, remain open. This note is a status brief with diagrams, not a submission claiming RH.

1 Setup

Fix a logarithmic window length $T > 0$ and set $x = e^{uT}$ for $u \in [0, 1]$. Let $\{\varphi_k\}_{k\geq 0}$ be the shifted Legendre orthonormal system on $L^2([0, 1], du)$:

$$\varphi_k(u) = \sqrt{2k+1}\, L_k(2u-1), \qquad V_d = \operatorname{span}\{\varphi_0, \ldots, \varphi_d\}, \quad P_d = \text{proj. onto } V_d.$$

For a residual $q \in L^2([0, 1])$ define the energy ratio

$$R_d(q) = \frac{\|P_d q\|_{L^2}^2}{\|q\|_{L^2}^2} = \frac{\sum_{k=0}^{d} |\langle q, \varphi_k\rangle|^2}{\|q\|^2} \in [0, 1], \qquad S_d(q; T) = T^{2(d+1)} R_d(q).$$

Low R_d: high-frequency content (RH-like for pure critical-line modes). High R_d: persistent low-degree mass (defect).

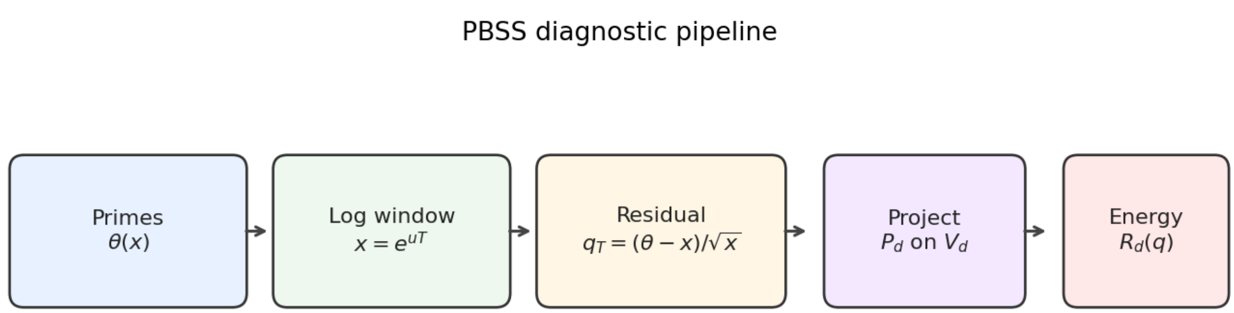

Figure 1: Diagnostic pipeline: Chebyshev residual on a log-window, projected onto low-degree Legendre modes; report R_d.

Model residuals. Critical-line (CL) mode $q_T^{\mathrm{cl}}(u) = \sin(tTu)$; off-critical $e^{T(\sigma-1/2)u}\sin(tTu)$; orthogonal defect $q = \sqrt{1-\varepsilon^2}\, f + \varepsilon\, \varphi_j$ with $f \perp V_d$, $\|f\|_2 = 1$.

Arithmetic residual (shipped probe). With $\theta(x) = \sum_{p \le x} \log p$,

$$q_T(u) = \mathrm{detrend}\Big(\frac{\theta(e^{uT}) - e^{uT}}{\sqrt{e^{uT}}}\Big) \quad \text{(default detrend: degree one).}$$

2 Proved diagnostic lemmas ($\mathbf{A}_0/\mathbf{B}_0$)

Lemma	Statement
M1	$R_d(\varphi_m) = 1_{m \le d}$
M2	Orthogonal defect $\Rightarrow R_d(q) = \varepsilon^2$
M3	CL pure mode $R_d(\sin(tTu)) = O_d(T^{-2})$ $(\mathbf{A}_0)$
M4	Fixed $\varepsilon > 0 \Rightarrow R_d \not\to 0$ (blocks vanishing)

M1–M4 are continuous L^2 arguments (`src/pbss/lemmas.py`) with discrete verification in `tests/`. They prove the *model* theorems:

- $\mathbf{A}_0$ **(proved):** critical-line pure mode has $R_d \to 0$ as $T \to \infty$.
- $\mathbf{B}_0$ **(proved):** a uniform low-degree defect keeps $R_d = \varepsilon^2 \not\to 0$.

Vanishing of R_d is *necessary* for the absence of a persistent polynomial defect. It is *not* a proof that arithmetic primes are free of such structure, nor a proof of RH.

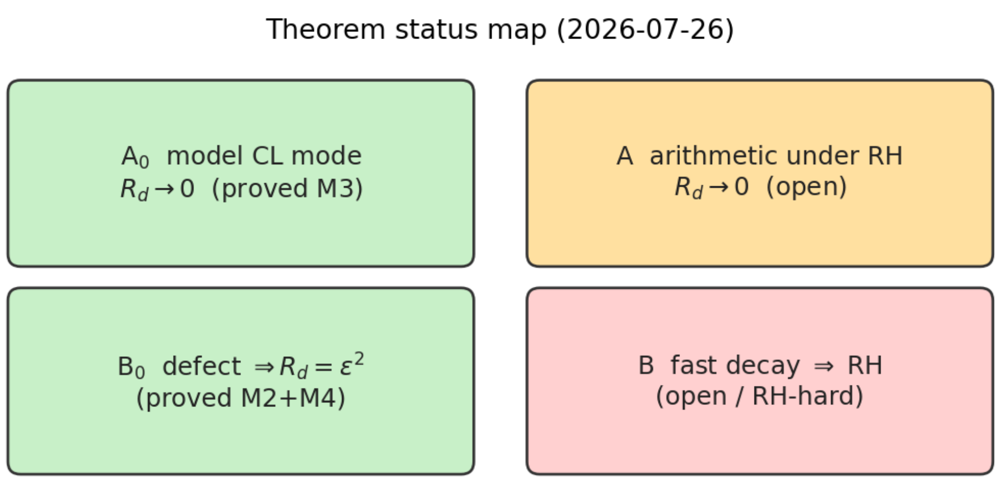

Figure 2: Status map. Green: proved model results. Amber/red: full A/B open.

3 Theorems A and B (open beyond models)

Full Theorem A (conditional on RH; open). Under RH, for the arithmetic residual q_T (or an equivalent explicit-formula residual), $R_d(q_T) \to 0$ as $T \to \infty$. Outline: RH $\Rightarrow$ superposition of CL modes; each contributes $O(T^{-2})$ by M3; controlling the sum over zeros is analytic number theory not completed here.

Full Theorem B (open; RH-hard). If $R_d(q_T)$ decays sufficiently fast, then ζ has no zero off the critical line. The implication is essentially as hard as RH; not proved in this repository.

4 Grand campaign numerics

Runner: `experiments/run_grand_campaign.py`. Scale: $x_{\max} = 10^{10}$ (455 052 511 primes, segmented sieve + on-disk checkpoint); 14 windows in T; ablations $d \in \{2, 4, 6, 8\}$, detrend $\in \{\text{none}, \text{deg0}, \text{deg1}\}$; controls CL / off-critical σ=0.75 / defect ε=0.5; **20 000** MC spectral-defect trials per T (280 000 total); wall $\sim$ 1 hour multi-core; resume-capable state JSON.

Table 1: Focus cut ($d = 4$, deg1) and controls at large T.

T	$x_{\max}$	R_4 arith	R_4 CL	R_4 defect	MC mean
10	$2{\times}10^4$	0.144	$\sim 6{\times}10^{-3}$	0.250	0.793
16	$9{\times}10^6$	0.193	$\sim 2{\times}10^{-3}$	0.250	0.793
23	$1{\times}10^{10}$	**0.155**	$\sim 5{\times}10^{-4}$	0.250	0.795

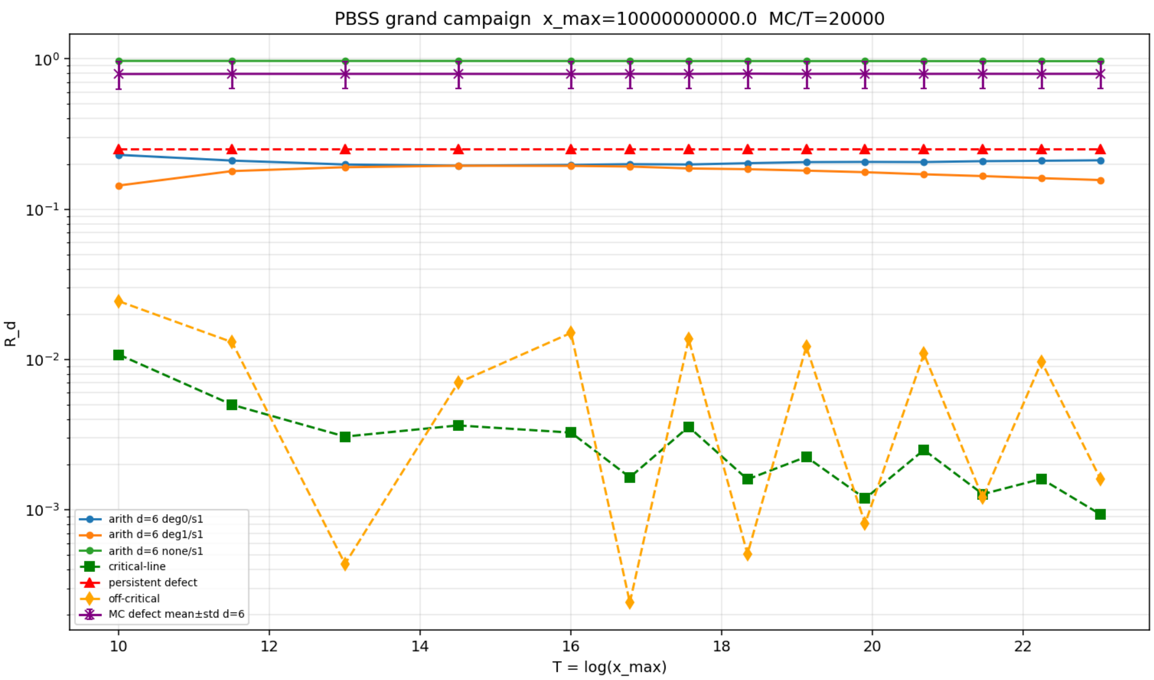

Figure 3: Grand campaign: arithmetic ablations, CL/defect/off-critical controls, MC defect mean $\pm$ std. Arithmetic tracks stay $O(10^{-1})$; CL decays; defects stay high.

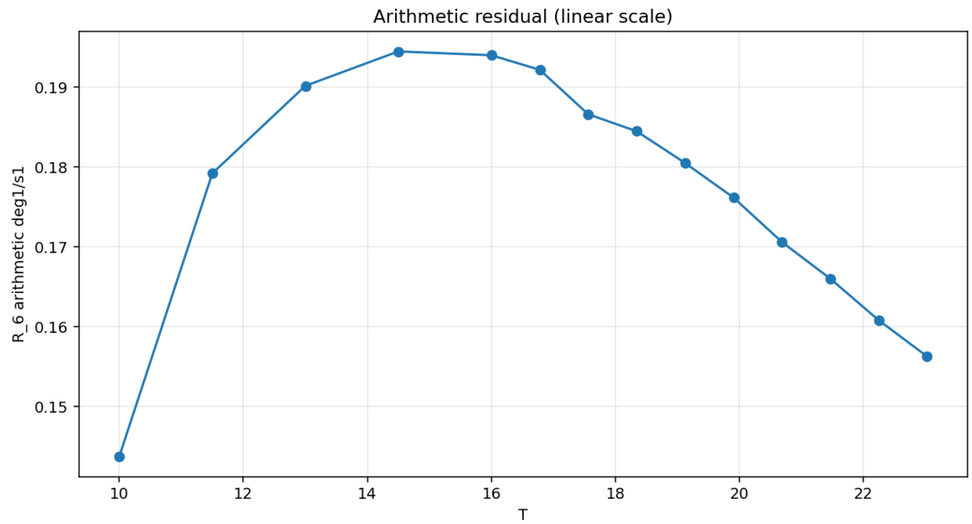

Figure 4: Arithmetic focus on a linear R_d scale: soft plateau ~ 0.15–0.19 through $T \approx 23$ ($x_{\max} = 10^{10}$).

Reading.

1. The diagnostic separates pure CL modes from planted defects as $\mathrm{A}_0/\mathrm{B}_0$ predict.
2. Arithmetic $q_T = (\theta - x)/\sqrt{x}$ (detrended) does **not** collapse like A_0 through ten billion; it soft-plateaus.
3. This does **not** prove or disprove RH. The current arithmetic probe is not yet the object for which full Theorem A is numerically visible.

5 Done vs open; next step

Done	**Open**
Lemmas M1–M4; $\mathrm{A}_0/\mathrm{B}_0$	Full Theorem A (arithmetic under RH)
Sieve + checkpoint to 10^{10}	Full Theorem B
Grand multi-T ablations, MC	Explicit-formula residual / zero-peeling
Optional CuPy (NumPy fallback)	Finite-mode sum lemma; stronger whitening
Tests green; docs locked	Legacy $P \approx 3.92$ normalization

Highest-leverage next step. Build an *explicit-formula residual*: subtract truncated zero sums / secondary main terms, then re-project. Couple with a proved finite-superposition form of M3. Larger x alone mostly reconfirms the plateau; the A-gap is the missing bridge from arithmetic q_T to CL mode sums.

6 Explicit non-claim

> *This repository and this note do not contain an unconditional proof of the Riemann Hypothesis. Proved results are about the diagnostic ($\mathrm{A}_0/\mathrm{B}_0$). Full A/B and RH remain open.*

Artifacts: `docs/STATUS.md`, `THEOREMS_AB.md`, `PROOFS_LEMMAS.md`, `results/grand_campaign/`. Private research; not for public distribution without permission.